American International University Bangladesh

Computer Graphics

Course Code: CSC4118

Fall Semester 2024-25

Faculty: Dipta Justin Gomes

Lecturer

Department of Computer Science, FST

TITLE: WASHINGTON CITY VIEW

Group Member Name	ID	Contribution
1. Jayed Hasan (Corresponding member)	22-46364-1	Day/Night Scene Development, River and Weather Effects, Animations (Cars, Sun, Moon), User Interaction Features
2. Dipra Kar	22-46336-1	Winter Scene Development, Animations (Snowfall, Boats), Building and Bridge Design
Section:	**G**	

Introduction

The Washington City View project uses OpenGL to create a realistic city environment in a virtual simulation of Washington, DC. With interactive and animated features like moving cars, boats, and weather effects like snowfall, the project offers three distinct scenarios: day, night, and winter. The goal of this project is to demonstrate how computer graphics can be used to produce realistic simulations for entertainment and urban visualization. Through the integration of technical accuracy and imaginative design, the project showcases the capabilities of modern graphical tools and methods.

Environment and Methods

This project has been developed using:

- **Environment**: OpenGL library in the C++ programming language, integrated with the GLUT toolkit. The development is carried out using the Code Blocks IDE on Windows OS.
- **Methods and Tools**: Geometric primitives, vector graphics, and advanced OpenGL functionalities like transformations and color blending. The code is modularized into functions for ease of design and maintenance.

Feature Set

The project includes the following features:

1. Dynamic day and night transitions. (Jayed)

2. Lifelike river rendering with distinct day and night variations. (Jayed)

3. Seasonal views: Winter (snowfall effects). (Dipra)

4. Detailed city skyline with vector-based buildings and distinct textures for day and night. (Dipra & Jayed)

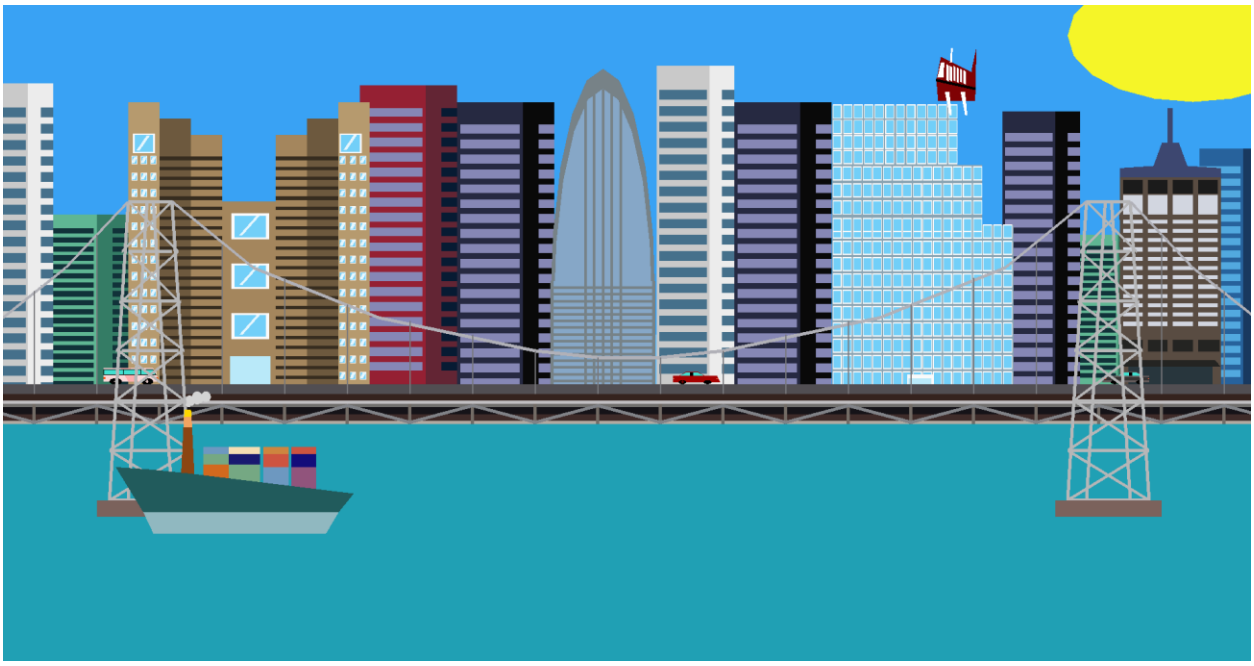
5. Moving vehicles (cars) with headlights at night on a detailed Washington Bridge. (Jayed)

6. Boats navigating the river (Dipra)
7. Sun and moon rendering based on the time of day (Jayed)
8. Snowfall animations with varying sizes and intensities for winter scenes. (Dipra)
9. OpenGL transformations for smooth and realistic object movements. (Dipra)
10. Interactive keyboard controls for scenario switching and animations. (Jayed)
11. Minimal lag for real-time rendering and efficient memory utilization. (Dipra)
12. Seamless transitions between different weather and time scenarios. (Dipra)

Conclusion

The Washington City View project is a prime example of the technical and artistic potential of computer graphics. With its dynamic weather, day-night transitions, and captivating animations, it accurately represents Washington, DC. Future improvements, like the addition of new landmarks or the integration of real-time data for adaptive weather patterns, are made possible by the project's modularity. Despite drawbacks such as hardware dependence, the project offers a strong foundation for investigating the possibilities of graphical simulations in the fields of entertainment, education, and urban planning. The team's completion of this project has given us invaluable experience using OpenGL for practical purposes.

Screenshots:



Day View

Night View

Winter View